



National  
Qualifications  
2016

**X757/75/02**

**Physics**  
**Section 1 — Questions**

TUESDAY, 24 MAY

1:00 PM – 3:00 PM

Instructions for the completion of Section 1 are given on *Page 02* of your question and answer booklet X757/75/01.

Record your answers on the answer grid on *Page 03* of your question and answer booklet

Reference may be made to the Data Sheet on *Page 02* of this booklet and to the Relationships Sheet X757/75/11.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



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## DATA SHEET

### Speed of light in materials

| Material       | Speed in $\text{m s}^{-1}$ |
|----------------|----------------------------|
| Air            | $3.0 \times 10^8$          |
| Carbon dioxide | $3.0 \times 10^8$          |
| Diamond        | $1.2 \times 10^8$          |
| Glass          | $2.0 \times 10^8$          |
| Glycerol       | $2.1 \times 10^8$          |
| Water          | $2.3 \times 10^8$          |

### Gravitational field strengths

|         | Gravitational field strength on the surface in $\text{N kg}^{-1}$ |
|---------|-------------------------------------------------------------------|
| Earth   | 9.8                                                               |
| Jupiter | 23                                                                |
| Mars    | 3.7                                                               |
| Mercury | 3.7                                                               |
| Moon    | 1.6                                                               |
| Neptune | 11                                                                |
| Saturn  | 9.0                                                               |
| Sun     | 270                                                               |
| Uranus  | 8.7                                                               |
| Venus   | 8.9                                                               |

### Specific latent heat of fusion of materials

| Material       | Specific latent heat of fusion in $\text{J kg}^{-1}$ |
|----------------|------------------------------------------------------|
| Alcohol        | $0.99 \times 10^5$                                   |
| Aluminium      | $3.95 \times 10^5$                                   |
| Carbon Dioxide | $1.80 \times 10^5$                                   |
| Copper         | $2.05 \times 10^5$                                   |
| Iron           | $2.67 \times 10^5$                                   |
| Lead           | $0.25 \times 10^5$                                   |
| Water          | $3.34 \times 10^5$                                   |

### Specific latent heat of vaporisation of materials

| Material       | Specific latent heat of vaporisation in $\text{J kg}^{-1}$ |
|----------------|------------------------------------------------------------|
| Alcohol        | $11.2 \times 10^5$                                         |
| Carbon Dioxide | $3.77 \times 10^5$                                         |
| Glycerol       | $8.30 \times 10^5$                                         |
| Turpentine     | $2.90 \times 10^5$                                         |
| Water          | $22.6 \times 10^5$                                         |

### Speed of sound in materials

| Material       | Speed in $\text{m s}^{-1}$ |
|----------------|----------------------------|
| Aluminium      | 5200                       |
| Air            | 340                        |
| Bone           | 4100                       |
| Carbon dioxide | 270                        |
| Glycerol       | 1900                       |
| Muscle         | 1600                       |
| Steel          | 5200                       |
| Tissue         | 1500                       |
| Water          | 1500                       |

### Specific heat capacity of materials

| Material  | Specific heat capacity in $\text{J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$ |
|-----------|---------------------------------------------------------------------------|
| Alcohol   | 2350                                                                      |
| Aluminium | 902                                                                       |
| Copper    | 386                                                                       |
| Glass     | 500                                                                       |
| Ice       | 2100                                                                      |
| Iron      | 480                                                                       |
| Lead      | 128                                                                       |
| Oil       | 2130                                                                      |
| Water     | 4180                                                                      |

### Melting and boiling points of materials

| Material  | Melting point in $^\circ\text{C}$ | Boiling point in $^\circ\text{C}$ |
|-----------|-----------------------------------|-----------------------------------|
| Alcohol   | -98                               | 65                                |
| Aluminium | 660                               | 2470                              |
| Copper    | 1077                              | 2567                              |
| Glycerol  | 18                                | 290                               |
| Lead      | 328                               | 1737                              |
| Iron      | 1537                              | 2737                              |

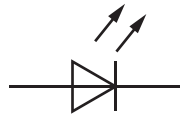
### Radiation weighting factors

| Type of radiation | Radiation weighting factor |
|-------------------|----------------------------|
| alpha             | 20                         |
| beta              | 1                          |
| fast neutrons     | 10                         |
| gamma             | 1                          |
| slow neutrons     | 3                          |
| X-rays            | 1                          |

## SECTION 1

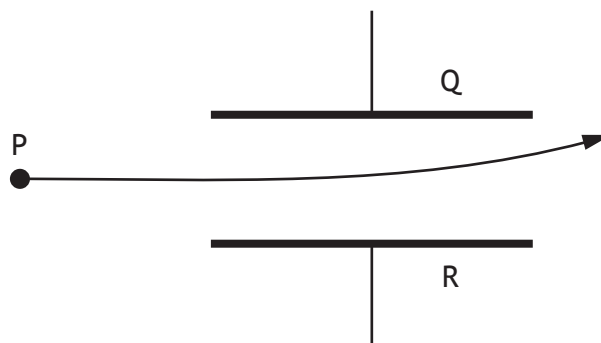
Attempt ALL questions

1. The symbol for an electronic component is shown.



This is the symbol for

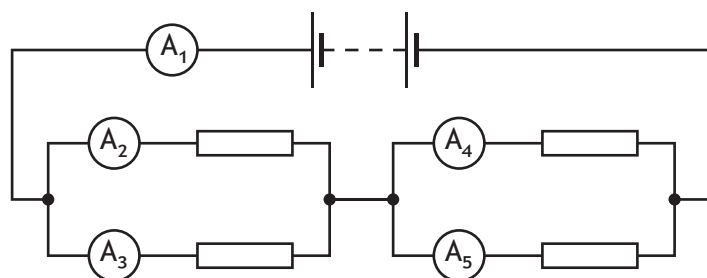
- A an LDR
  - B a transistor
  - C an LED
  - D a photovoltaic cell
  - E a thermistor.
2. A uniform electric field exists between plates Q and R.  
The diagram shows the path taken by a particle as it passes through the field.



Which row in the table identifies the charge on the particle, the charge on plate Q and the charge on plate R?

|   | <i>Charge on particle</i> | <i>Charge on plate Q</i> | <i>Charge on plate R</i> |
|---|---------------------------|--------------------------|--------------------------|
| A | negative                  | positive                 | negative                 |
| B | negative                  | negative                 | positive                 |
| C | no charge                 | negative                 | positive                 |
| D | no charge                 | positive                 | negative                 |
| E | positive                  | positive                 | negative                 |

3. A circuit is set up as shown.



The reading on ammeter  $A_1$  is  $5.0\text{ A}$ .

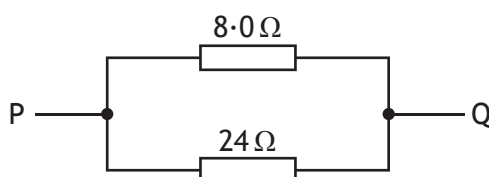
The reading on ammeter  $A_2$  is  $2.0\text{ A}$ .

The reading on ammeter  $A_4$  is  $1.0\text{ A}$ .

Which row in the table shows the reading on ammeters  $A_3$  and  $A_5$ ?

|   | Reading on ammeter $A_3$<br>(A) | Reading on ammeter $A_5$<br>(A) |
|---|---------------------------------|---------------------------------|
| A | 2.0                             | 1.0                             |
| B | 3.0                             | 1.0                             |
| C | 2.0                             | 4.0                             |
| D | 3.0                             | 4.0                             |
| E | 5.0                             | 5.0                             |

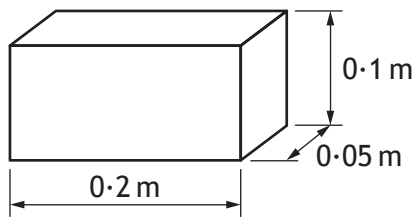
4. Two resistors are connected as shown.



The total resistance between P and Q is

- A  $0.17\ \Omega$
- B  $3.0\ \Omega$
- C  $6.0\ \Omega$
- D  $16\ \Omega$
- E  $32\ \Omega$ .

5. A block has the dimensions shown.

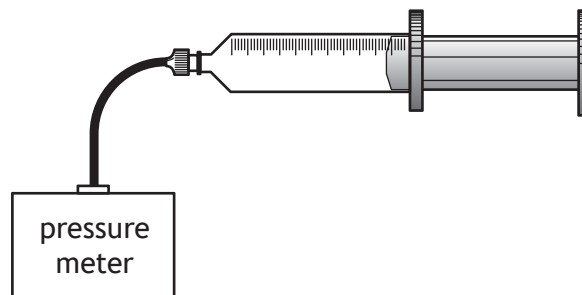


The block is placed so that one of the surfaces is in contact with a smooth table top.

The weight of the block is 4.90 N.

The **minimum** pressure exerted by the block on the table top is

- A 25 Pa
  - B 245 Pa
  - C 490 Pa
  - D 980 Pa
  - E 4900 Pa.
6. A syringe is connected to a pressure meter as shown.



The syringe contains a fixed mass of air of volume  $150 \text{ mm}^3$ .

The reading on the pressure meter is 120 kPa.

The volume of air inside the syringe is now changed to  $100 \text{ mm}^3$ .

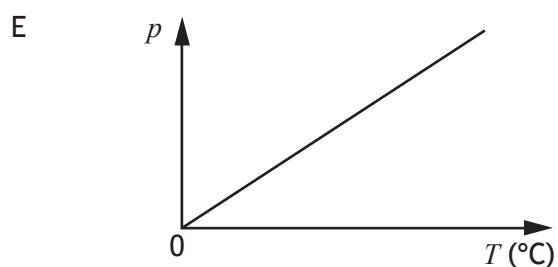
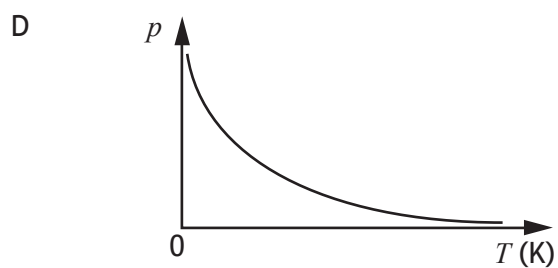
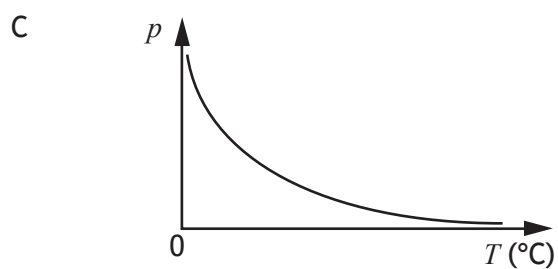
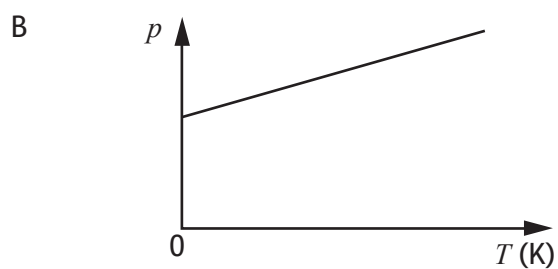
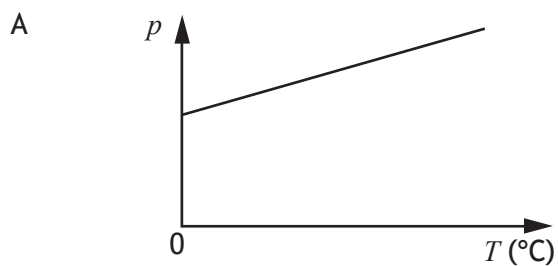
The temperature of the air in the syringe remains constant.

The reading on the pressure meter is now

- A 80 kPa
- B 125 kPa
- C 180 kPa
- D 80 000 kPa
- E 180 000 kPa.

7. A sample of an ideal gas is enclosed in a sealed container.

Which graph shows how the pressure  $p$  of the gas varies with the temperature  $T$  of the gas?



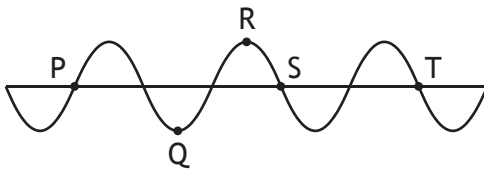
8. A student makes the following statements about waves.

- I Waves transfer energy.
- II A wave with a short wavelength diffracts more than a wave with a long wavelength.
- III The amplitude of a wave depends on its wavelength.

Which of these statements is/are correct?

- A I only
- B II only
- C III only
- D I and II only
- E I and III only

9. The diagram represents a wave.

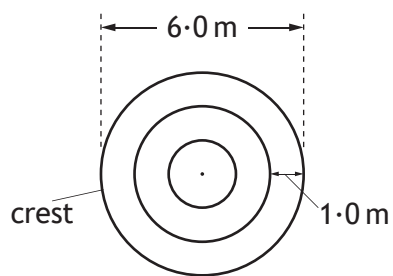


The wavelength of the wave is the horizontal distance between points

- A P and Q
- B P and S
- C Q and R
- D R and S
- E S and T.

[Turn over

10. The diagram represents the position of the crests of waves 3 seconds after a stone is thrown into a pool of still water.

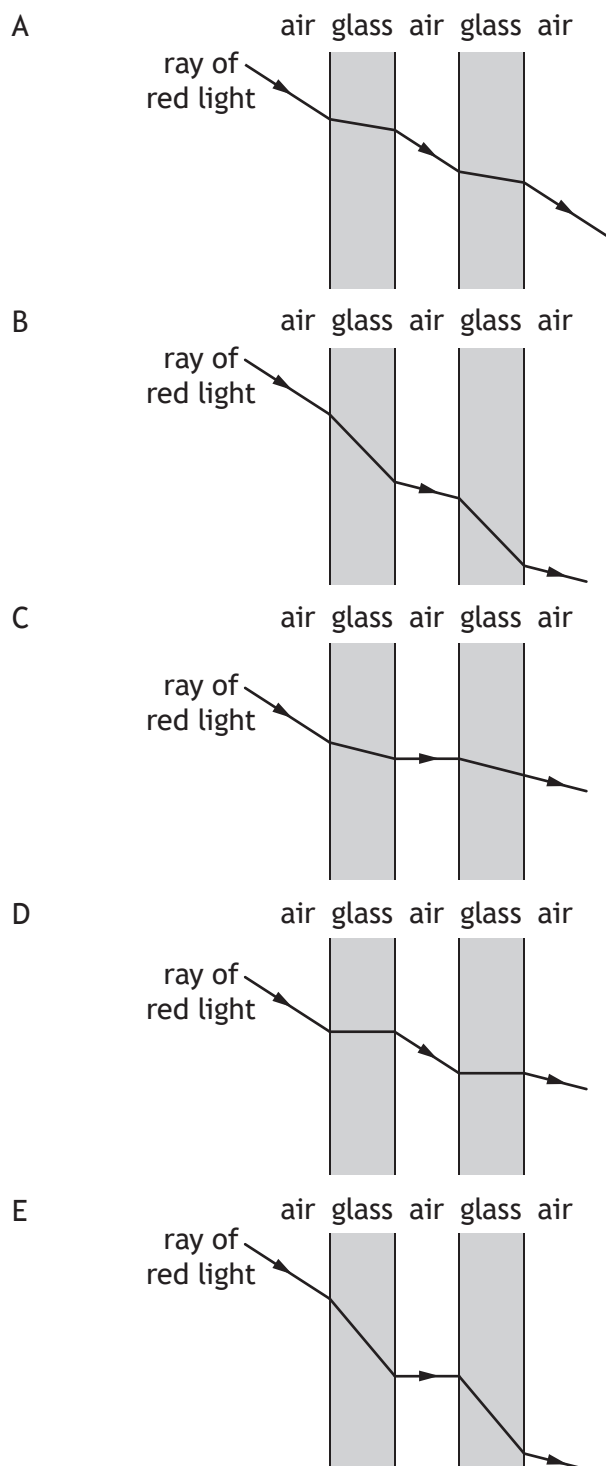


Which row in the table shows the speed and the frequency of the waves?

|   | <i>Speed</i><br>(m s <sup>-1</sup> ) | <i>Frequency</i><br>(Hz) |
|---|--------------------------------------|--------------------------|
| A | 0.33                                 | 3                        |
| B | 0.33                                 | 1                        |
| C | 1.0                                  | 1                        |
| D | 1.0                                  | 3                        |
| E | 1.0                                  | 4                        |



11. A ray of red light passes through a double glazed window.  
Which diagram shows the path of the ray as it passes through the window?



[Turn over

12. Which row in the table shows how the mass and charge of an alpha particle compares to the mass and charge of a beta particle?

|   | <i>Mass of an alpha particle compared to mass of a beta particle</i> | <i>Charge on an alpha particle compared to charge on a beta particle</i> |
|---|----------------------------------------------------------------------|--------------------------------------------------------------------------|
| A | larger                                                               | same                                                                     |
| B | larger                                                               | opposite                                                                 |
| C | same                                                                 | same                                                                     |
| D | smaller                                                              | opposite                                                                 |
| E | smaller                                                              | same                                                                     |

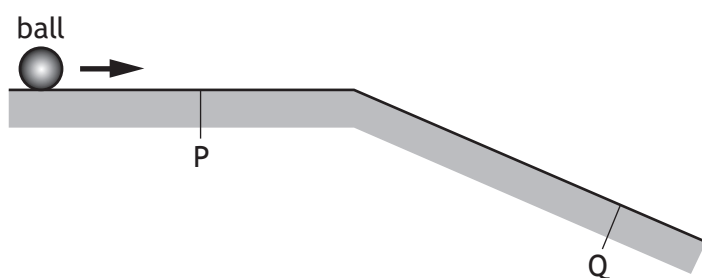
13. During ionisation an atom becomes a positive ion.  
Which of the following has been removed from the atom?

- A An alpha particle
- B An electron
- C A gamma ray
- D A neutron
- E A proton

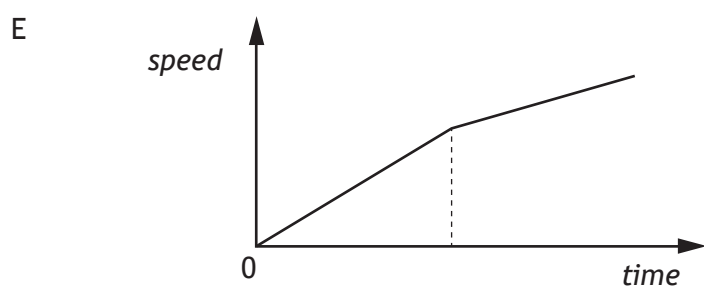
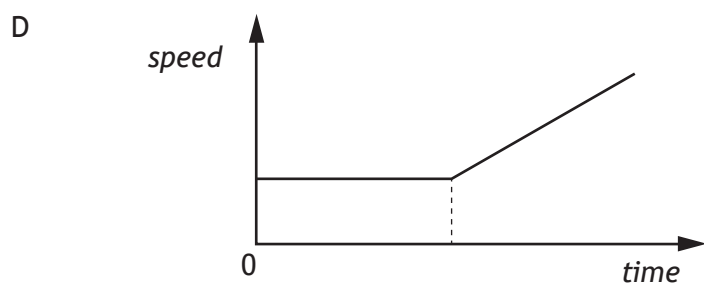
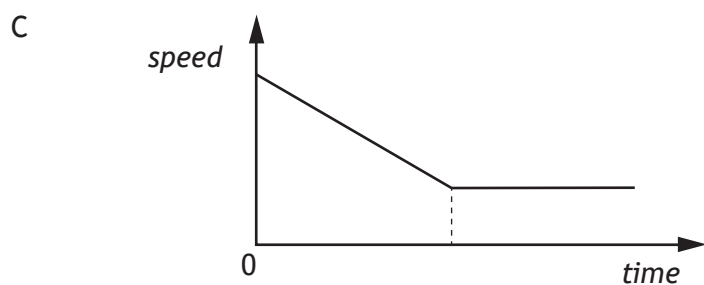
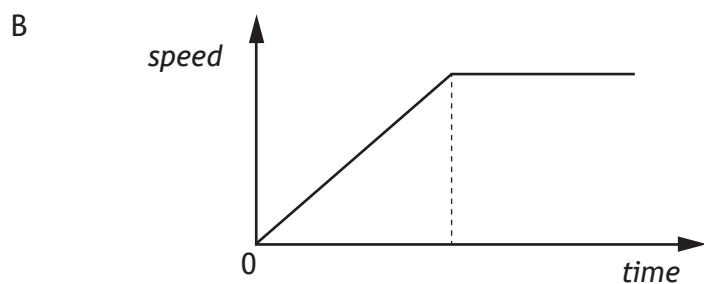
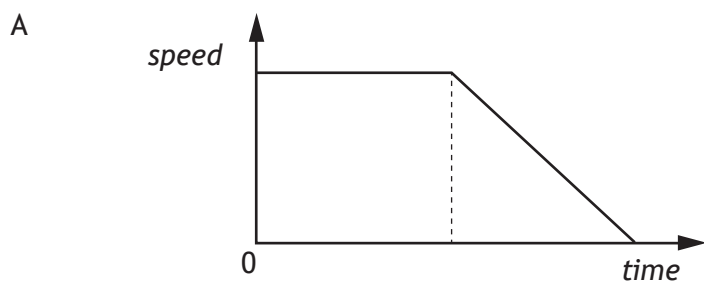
14. Which of the following is a vector quantity?

- A Mass
- B Time
- C Speed
- D Kinetic energy
- E Acceleration

15. A ball moves along a horizontal frictionless surface and down a slope as shown.

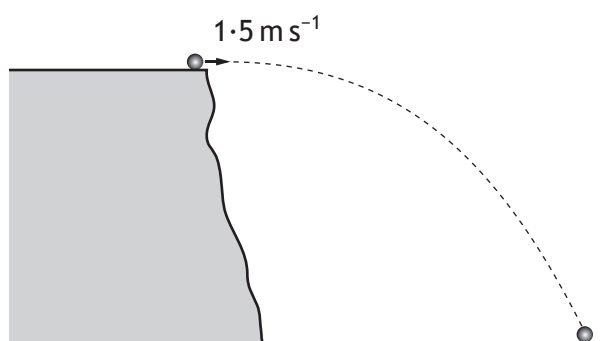


Which of the following graphs shows how the speed of the ball varies with time as it travels from P to Q?



16. A cyclist is travelling at  $10 \text{ m s}^{-1}$  along a level road.  
The cyclist applies the brakes and comes to rest in a time of 5 s.  
The combined mass of the cycle and cyclist is 80 kg.  
The maximum energy converted to heat by the brakes is
- A 160 J
  - B 400 J
  - C 800 J
  - D 4000 J
  - E 8000 J.
17. A rocket is taking off from the surface of the Earth. The rocket engines exert a force on the exhaust gases.  
Which of the following is the reaction to this force?
- A The force of the Earth on the exhaust gases.
  - B The force of the Earth on the rocket engines.
  - C The force of the rocket engines on the Earth.
  - D The force of the exhaust gases on the Earth.
  - E The force of the exhaust gases on the rocket engines.

18. A ball is projected horizontally with a velocity of  $1.5 \text{ m s}^{-1}$  from a cliff as shown.



The ball hits the ground  $1.2 \text{ s}$  after it leaves the cliff.

The effects of air resistance are negligible.

Which row in the table shows the horizontal velocity and vertical velocity of the ball just before it hits the ground?

|   | Horizontal velocity<br>( $\text{m s}^{-1}$ ) | Vertical velocity<br>( $\text{m s}^{-1}$ ) |
|---|----------------------------------------------|--------------------------------------------|
| A | 12                                           | 12                                         |
| B | 12                                           | 1.5                                        |
| C | 1.5                                          | 12                                         |
| D | 1.5                                          | 13                                         |
| E | 0                                            | 12                                         |

19. The minimum amount of energy required to change  $0.5 \text{ kg}$  of water at its boiling point into steam at the same temperature is

- A  $2.09 \times 10^3 \text{ J}$
- B  $1.67 \times 10^5 \text{ J}$
- C  $3.34 \times 10^5 \text{ J}$
- D  $1.13 \times 10^6 \text{ J}$
- E  $2.26 \times 10^6 \text{ J}$ .

20. A student makes the following statements about the Universe.

- I The Big Bang Theory is a theory about the origin of the Universe.
- II The Universe is approximately 14 million years old.
- III The Universe is expanding.

Which of these statements is/are correct?

- A I only
- B II only
- C I and II only
- D I and III only
- E I, II and III.

**[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF  
YOUR QUESTION AND ANSWER BOOKLET]**

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